

LSGI3242A Digital Terrain Modelling Lab Assignment 2

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Procedures for processing DTM data

1. Find the N (number of items) (**Appendix A Fig. 2a**)
2. Find the M (average value of items in the objective area) (**Appendix A Fig. 2b**)
3. Find the intermediate values for Cov(1) (Covariance of objective area and area offset by d columns). For each value in the objective area, find the quantity of difference of the values and the average M, and the quantity of difference of the extended area, which is offset by d columns, and multiply the two quantities. (**Appendix A Fig. 1a**)
4. Sum all intermediate values in step 3, then divide by N-1 to obtain Cov(1) (**Appendix A Fig. 2c**)
5. Find the intermediate values for V(0) (Variance of objective area using the objective area's mean). For each value in the objective area, find the quantity of difference of the objective area and the average M, squared. (**Appendix A Fig. 1b**)
6. Sum all intermediate values in step 5, then divide by N-1 to obtain V(0) (**Appendix A Fig. 2d**)
7. Find the intermediate values for V(1) (Variance of area offset by 1 column using the objective area's mean). For each value in the extended area, which is offset by d columns, find the quantity of difference of the values and the average M, squared. (**Appendix A Fig. 1b**)
8. Sum all intermediate values in step 7, then divide by N-1 to obtain V(1) (**Appendix A Fig. 2e**)
9. The reason for Cov(d) and V(d) divides sum of intermediate values by N-1 is because they are samples from the whole dataset, which compensates the bias from without the whole population data and mean.
10. Obtain R(d) with $\text{Cov}(d) / \sqrt{V(0) * V(d)}$ (**Appendix A Fig. 2f**)
11. Repeat the steps 1-9 with d = 2 to 8. Values N, M, and V(0) are already found.
12. Copy the R(d) values (**Appendix A Fig. 3a, 3b**)
13. Make a table with d values and R(d) values, and plot a table (**Appendix A Fig. 4**).

Results

The plot (**Appendix A Fig. 4**) appears to show that as the interval d increases from 1 to 8, the correlation coefficient R(d) decreases from 0.90 to 0.02. In this assignment, the dataset shows a negative correlation between interval and correlation coefficient. This also appears to show The First Law of Geography, according to Waldo Tobler, "everything is related to everything else, but near things are more related than distant things." (Tobler, 1970, p. 236), as the more columns deviate from the objective area, the correlation coefficient decreases.

Appendix A

Screenshots of procedures for processing data

Formula bar: $=(A1-\$B\$19)*(B1-\$B\$19)$

	A	B	C	D	E	F
	$(Z_i - M)(Z_{i+1} - M)$					
	691.5947	573.7197	262.2197	316.0947	-200.218	-174.624
	5.844727	143.001	-17.5615	-5.31152	-22.6553	-94.499
	-53.874	-12.249	1.875977	15.15723	43.68848	14.59473
	132.501	166.626	498.6572	371.001	211.1572	254.751
	565.251	811.626	1760.376	1315.063	1196.938	1278.001

Fig. 1a: Formula for each intermediate item for $Cov(1)$

Formula bar: $=(A1-\$B\$19)^2$

	A	B	C	D	E	F	G
	$(Z_i - M)^2$						
	550.7822	868.4072	379.0322	181.4072	550.7822	72.78223	418.9697
	0.219727	155.4697	131.5322	2.344727	12.03223	42.65723	209.3447
	241.2197	12.03223	12.46973	0.282227	814.0322	2.344727	90.84473
	241.2197	72.78223	381.4697	651.8447	211.1572	211.1572	307.3447
	757.9697	421.5322	1562.72	1983.032	872.0947	1642.782	994.2197

Fig. 1b: Formula for each intermediate item for $V(0)$

Formula bar: $=(B1-\$B\$19)^2$

	A	B	C	D	E	F
	$(Z_{i+1} - M)^2$					
	868.4072	379.0322	181.4072	550.7822	72.78223	418.96
	155.4697	131.5322	2.344727	12.03223	42.65723	209.34
	12.03223	12.46973	0.282227	814.0322	2.344727	90.844
	72.78223	381.4697	651.8447	211.1572	211.1572	307.34
	421.5322	1562.72	1983.032	872.0947	1642.782	994.21
	1262.47	4426.407	1408.595	2759.532	2453.345	183.09
	3083.72	8377.97	4294.345	5704.97	3425.907	2.3447

Fig. 1c: Formula for each intermediate item for $V(1)$

A19		=COUNTA(A1:P16)						
	A	B	C	D	E	F	G	H
7								
8	N	M	Cov(1)	V(0)	V(1)	R(1)		
9	256	1840.531	10321.38	10740.41	12205.6	0.901462		
20								

Fig. 2a: Formula for N

B19		=AVERAGE(A1:P16)					
	A	B	C	D	E	F	G
17							
18	N	M	Cov(1)	V(0)	V(1)	R(1)	
19	256	1840.531	10321.38	10740.41	12205.6	0.901462	
20							

Fig. 2b: Formula for M

C19		=SUM(A22:P37)/(A19-1)					
	A	B	C	D	E	F	G
17							
18	N	M	Cov(1)	V(0)	V(1)	R(1)	
19	256	1840.531	10321.38	10740.41	12205.6	0.901462	
20							

Fig. 2c: Formula for $Cov(1)$

D19		=SUM(A40:P55)/(A19-1)					
	A	B	C	D	E	F	G
17							
18	N	M	Cov(1)	V(0)	V(1)	R(1)	
19	256	1840.531	10321.38	10740.41	12205.6	0.901462	
20							

Fig. 2d: Formula for $V(0)$

E19		=SUM(A58:P73)/(A19-1)					
	A	B	C	D	E	F	G
17							
18	N	M	Cov(1)	V(0)	V(1)	R(1)	
19	256	1840.531	10321.38	10740.41	12205.6	0.901462	
20							

Fig. 2e: Formula for $V(1)$

F19		=C19/SQRT(D19*E19)					
	A	B	C	D	E	F	G
17							
18	N	M	Cov(1)	V(0)	V(1)	R(1)	
19	256	1840.531	10321.38	10740.41	12205.6	0.901462	
20							

Fig. 2f: Formula for $R(1)$

17						
18	N	M	Cov(1)	V(0)	V(1)	R(1)
19	256	1840.531	10321.38	10740.41	12205.6	0.901462
20						

Fig. 3a: Results of N , M , $Cov(1)$, $V(0)$, $V(1)$, and $R(1)$

17						
18	N	M	Cov(2)	V(0)	V(2)	R(2)
19	256	1840.531	8650.571	10740.41	12863.99	0.735947
20						

Fig. 3b: Results of N , M , $Cov(2)$, $V(0)$, $V(2)$, and $R(2)$

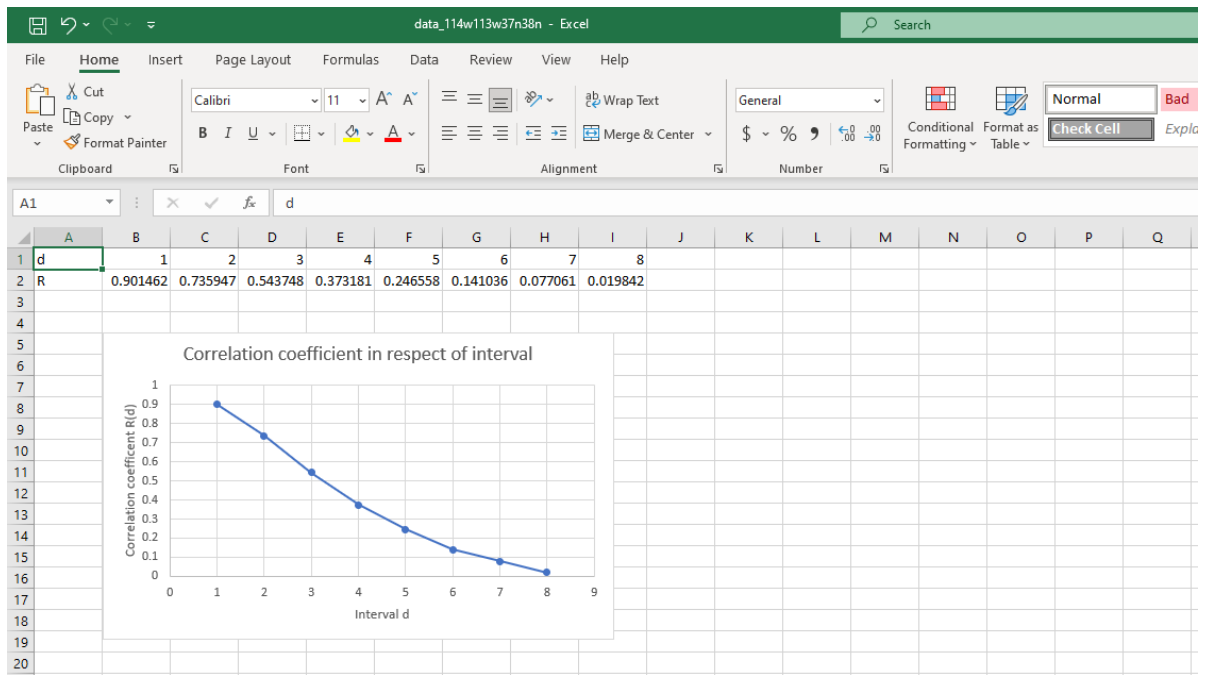


Fig. 4: Plot showing relationship of $R(d)$ (correlation coefficient) in respect of d (interval)

Appendix B

Selected 16x24 raw data

1864	1870	1860	1854	1864	1832	1861	1871	1887	1910	1955	1962	1958	1954	1979	2008	2072	2061	2067	1993	1913	1872	1849	1834
1841	1853	1852	1839	1844	1834	1855	1860	1902	1880	1897	1909	1936	1980	2011	2095	2172	2100	1964	1927	1938	1841	1824	1812
1825	1844	1837	1840	1812	1839	1831	1868	1854	1876	1903	1933	1949	1980	2055	2154	2218	2059	1921	1877	1836	1807	1801	1793
1825	1832	1821	1815	1826	1826	1823	1833	1850	1881	1903	1931	1978	2029	2259	2187	2092	2001	1924	1838	1810	1790	1780	1773
1813	1820	1801	1796	1811	1800	1809	1870	1862	1877	1912	1952	1994	2046	2235	2150	2047	1931	1845	1803	1782	1768	1764	1758
1788	1805	1774	1803	1788	1791	1827	1831	1850	1886	1916	1943	1987	2080	2152	2011	2013	1864	1807	1774	1759	1753	1751	1745
1776	1785	1749	1775	1765	1782	1839	1816	1849	1886	1916	1945	2004	2127	2082	1973	1878	1823	1785	1753	1745	1742	1739	1742
1757	1738	1762	1747	1769	1787	1815	1848	1839	1871	1896	1898	1942	2097	2021	1883	1812	1782	1750	1736	1736	1736	1745	1738
1733	1743	1739	1748	1772	1803	1839	1849	1888	1879	1870	1856	1893	1888	1856	1820	1775	1745	1736	1739	1742	1739	1737	1737
1723	1725	1726	1753	1785	1823	1857	1931	1889	1832	1835	1809	1807	1806	1785	1752	1736	1737	1743	1743	1744	1744	1742	1742
1708	1707	1733	1773	1798	1852	1956	1947	1852	1818	1776	1779	1771	1755	1738	1736	1734	1736	1737	1740	1740	1742	1744	1745
1696	1697	1722	1772	1841	1907	1960	1888	1811	1770	1751	1749	1743	1735	1736	1741	1739	1739	1741	1743	1744	1742	1746	1748
1682	1771	1714	1713	1776	1859	1978	1856	1759	1736	1736	1736	1736	1736	1737	1741	1743	1746	1743	1747	1747	1749	1749	1751
1689	1718	1747	1804	1883	1820	1755	1755	1736	1736	1739	1736	1736	1738	1739	1744	1745	1747	1751	1751	1754	1756	1758	1758
1731	1765	1768	1850	1932	1880	1851	1747	1736	1745	1738	1737	1740	1739	1744	1748	1750	1757	1760	1761	1764	1764	1765	1763
1799	1792	1781	1876	1974	1955	1841	1771	1750	1737	1738	1739	1742	1745	1747	1750	1754	1764	1769	1774	1773	1774	1774	1771

References

Tobler, W. R. 1970. A computer movie simulating urban growth in the Detroit region. *Economic Geography*. 46, (1970), 236.